STUDENT ENROLLMENT & COURSE MANAGEMENT SYSTEM

PROJECT DEVELOPMENT DOCUMENTATION

PROJECT OVERVIEW

The Student Enrollment & Course Management System is designed to manage student details, courses, enrollment records, and academic information in an organized manner. The system uses Salesforce features such as Objects, Fields, Relationships, Validation Rules, Lightning App, Page Layouts, Flow Automation, Reports, Dashboards, and Security.

The project is developed in five phases, starting from requirements analysis and planning and ending with deployment, documentation, and final presentation.

PHASE 1 – REQUIREMENTS ANALYSIS & PLANNING

Main Topic: Requirements Analysis & Planning

Requirements analysis is the first phase of the project. In this phase, the requirements of students, administrators, and course management are identified.

Objectives

- Manage student information.
- Manage course information.
- Allow students to enroll in courses.
- Maintain enrollment records.
- Reduce manual data entry.
- Provide reports and dashboards.
- Maintain data security.

Sub Topic: Creating Development & Configuration

The development environment is prepared before starting the implementation.

- Create and configure the Salesforce development environment.
- Identify the required objects and fields.
- Plan relationships between objects.
- Configure user access and permissions.
- Prepare the development environment for implementation.
- Define the overall project structure.

PHASE 2 – BACKEND DEVELOPMENT & CONFIGURATION

Main Topic: Backend Development & Configuration

In this phase, the database structure and backend components are created using Salesforce objects and configuration features.

1. Objects

Objects are used to store and manage project data.

- Student
- Course
- Enrollment

The Student Object stores student details. The Course Object stores course information. The Enrollment Object stores student enrollment details.

2. Fields

Fields are created to store specific information.

- Student: Student Name, Email, Phone, Date of Birth, Department.
- Course: Course Name, Course Code, Course Duration, Course Fee.
- Enrollment: Enrollment Date, Student, Course, Enrollment Status.

3. Relationships

Relationships connect different objects. A student can enroll in different courses, and each course can have multiple students. The basic relationship is Student → Enrollment → Course.

4. Validation Rules

Validation rules ensure that correct data is entered into the system.

- Email must be entered correctly.
- Phone number should contain valid digits.
- Course fee should not be negative.
- Required fields cannot be left empty.
- Enrollment date should contain a valid date.

5. Approval Process

The approval process is used to review and approve enrollment records.

Student Enrollment → Review → Approval → Enrollment Confirmed

This helps maintain proper control over the enrollment process.

PHASE 3 – UI/UX DEVELOPMENT & CUSTOMIZATION

Main Topic: UI/UX Development & Customization

This phase focuses on creating a simple and user-friendly interface for managing student and course information.

1. Tabs

Tabs provide quick access to different objects in the system.

- Students
- Courses
- Enrollments
- Reports
- Dashboards

2. Lightning App

A Lightning App is created to provide a centralized interface for the Student Enrollment & Course Management System. The app provides easy access to student, course, enrollment, reports, and dashboard information.

3. Page Layouts

Page layouts control the fields and sections displayed on record pages. The page layouts are customized for Student, Course, and Enrollment records. Required fields are displayed clearly to make data entry easier.

4. Record Pages

Record pages are customized to display important information in an organized manner. The Student record page can display student details, contact information, enrolled courses, and enrollment status.

PHASE 4 – DATA MIGRATION, TESTING & SECURITY

Main Topic: Data Migration, Testing & Security

This phase focuses on importing data, testing system functionality, automating processes, and securing the application.

1. Data Migration

Existing student and course data can be imported into Salesforce.

- Prepare the data.
- Check the data for errors.
- Import student and course records.
- Verify the imported records.
- Correct any errors.

2. Automation with Flow

Salesforce Flow is used to automate repetitive tasks.

Example: Enrollment Created → Flow Runs → Status Updated → Notification/Action

Flow reduces manual work and improves process efficiency.

3. Reports

Reports are used to analyze system data.

- Total Students
- Available Courses
- Student Enrollment Details
- Course-wise Enrollment
- Approved and Pending Enrollments

4. Dashboards

Dashboards provide a visual summary of important information.

- Total Students
- Total Courses
- Total Enrollments
- Pending Enrollments
- Course-wise Enrollment

Dashboards help administrators understand the current status of the system easily.

5. Security Setup

Security features are configured to protect student and course information.

- User permissions
- Profiles
- Permission Sets
- Role-based access
- Object-level security
- Field-level security
- Record-level access

Only authorized users can access or modify important information.

PHASE 5 – DEPLOYMENT, DOCUMENTATION & FINAL PRESENTATION

Main Topic: Deployment, Documentation & Final Presentation

In the final phase, the completed system is prepared for deployment and presentation.

Deployment

The configured components are checked and deployed to the required environment.

Documentation

The project documentation contains:

- Project Overview
- Requirements
- Objects and Fields
- Relationships
- Validation Rules
- Approval Process
- Lightning App
- Page Layouts
- Flow Automation
- Reports and Dashboards
- Security Setup
- Testing
- Conclusion

Final Presentation

The final presentation explains the purpose, development phases, features, and results of the project.

CONCLUSION

The Student Enrollment & Course Management System provides an organized way to manage students, courses, and enrollment records. Salesforce features such as Objects, Fields, Relationships, Validation Rules, Approval Processes, Lightning App, Page Layouts, Flow Automation, Reports, Dashboards, and Security help to create an efficient and user-friendly system.

The project reduces manual work, improves data management, provides better visibility through reports and dashboards, and protects important student information through security controls. Overall, the system provides a centralized platform for managing student enrollment and course-related activities efficiently.